

```
!pip install roboflow
!pip install ultralytics
```

```
Requirement already satisfied: opencv-python!=4.13.0.90,>=4.7.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.11.0+cu128)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (0.26.0+cu128)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (1.35.2)
Requirement already satisfied: nvidia-ml-py>=12.0.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (13.610.43)
Collecting ultralytics-thop>=2.1.6 (from ultralytics)
  Downloading ultralytics_thop-2.1.6-py3-none-any.whl.metadata (13 kB)
Collecting ultralytics-platform>=0.1.3 (from ultralytics)
  Downloading ultralytics_platform-0.1.11-py3-none-any.whl.metadata (8.4 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.2.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.10.0)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.53.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.4.5)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (24.1)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.1.3)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0)
Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.13/dist-packages (from polars>=0.20.0->ultralytics) (1.35.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultralytics) (3.3.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultralytics) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultralytics) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultralytics) (2024.7.4)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (4.12.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.4.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (2025.3.0)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (12.8.1)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (12.9.4)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.4.5)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.6.0)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (12.8.4.1)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (12.8.90)
Requirement already satisfied: nvidia-cuffrt-cu12==11.3.3.83.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (11.3.3.83)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (1.13.1.3)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (12.8.90)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (12.5.8.93)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (12.8.93)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (12.8.93)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart]>=12.8.1->ultralytics) (12.8.90)
Requirement already satisfied: httpx<1,>=0.28 in /usr/local/lib/python3.13/dist-packages (from ultralytics-platform>=0.1.3->ultralytics) (0.28.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.13/dist-packages (from cuda-bindings<13,>=12.9.4->ultralytics) (1.1.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.28->ultralytics-platform>=0.1.3->ultralytics) (4.6.2)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.28->ultralytics-platform>=0.1.3->ultralytics) (1.0.5)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1,>=0.28->ultralytics-platform>=0.1.3->ultralytics) (0.14.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from Jinja2->torch>=1.8.0->ultralytics) (2.1.5)
Downloading ultralytics-8.4.126-nv3-none-any.whl (1.4 MB)
```

```
from roboflow import Roboflow
rf = Roboflow(api_key="sjCv0N7rr3W19Hw2lgqJ")
project = rf.workspace("workspace-5ujvu").project("basketball-players-fy4c2-vfsuv")
version = project.version(17)
dataset = version.download("yolov5")
```

```
loading Roboflow workspace...
loading Roboflow project...
Downloading Dataset Version Zip in Basketball-Players-17 to yolov5pytorch:: 100%|██████████| 23029/23029 [00:02<00:00, 8376.41it/s]
Extracting Dataset Version Zip to Basketball-Players-17 in yolov5pytorch:: 100%|██████████| 652/652 [00:00<00:00, 7357.18it/s]
```

Double-click (or enter) to edit

```
import shutil
shutil.move("Basketball-Players-17/train"
            , "Basketball-Players-17/Basketball-Players-17/train")

shutil.move("Basketball-Players-17/valid"
            , "Basketball-Players-17/Basketball-Players-17/valid")
```

```
'Basketball-Players-17/Basketball-Players-17/valid'
```

```
!yolo task=detect mode=train model=yolov5l6u.pt data={dataset.location}/data.yaml epochs=100 imgsz=640 plots=True batch=8
```

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
83/100	6.9G	0.6369	0.4374	0.9425	179	640: 100% 32/32 2.6it/s 12.1s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.3it/s 0.6s
	all	32	483	0.971	0.907	0.943 0.767
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
84/100	6.9G	0.6244	0.4248	0.9352	204	640: 100% 32/32 2.6it/s 12.1s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.2it/s 0.6s
	all	32	483	0.972	0.906	0.943 0.762
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
85/100	6.9G	0.6366	0.4313	0.9489	175	640: 100% 32/32 2.7it/s 12.1s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.1it/s 0.7s
	all	32	483	0.955	0.911	0.944 0.76
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
86/100	6.9G	0.632	0.4264	0.9438	203	640: 100% 32/32 2.6it/s 12.1s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.1it/s 0.6s
	all	32	483	0.967	0.913	0.951 0.765
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
87/100	6.9G	0.627	0.4215	0.938	164	640: 100% 32/32 2.6it/s 12.1s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.2it/s 0.6s
	all	32	483	0.975	0.907	0.947 0.766
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
88/100	6.9G	0.6348	0.4326	0.9436	155	640: 100% 32/32 2.6it/s 12.2s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.2it/s 0.6s
	all	32	483	0.973	0.908	0.945 0.768
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
89/100	6.9G	0.6053	0.4126	0.9298	169	640: 100% 32/32 2.6it/s 12.3s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.2it/s 0.6s
	all	32	483	0.974	0.907	0.948 0.771
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
90/100	6.9G	0.5971	0.4069	0.9308	201	640: 100% 32/32 2.6it/s 12.1s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.2it/s 0.6s
	all	32	483	0.978	0.902	0.948 0.761
Closing dataloader mosaic						
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average')						
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
91/100	6.9G	0.564	0.3775	0.9074	95	640: 100% 32/32 2.5it/s 12.7s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.2it/s 0.6s
	all	32	483	0.981	0.897	0.949 0.764
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
92/100	6.9G	0.5468	0.3559	0.8958	115	640: 100% 32/32 2.7it/s 12.0s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.0it/s 0.7s
	all	32	483	0.972	0.902	0.951 0.77
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
93/100	6.9G	0.5365	0.3486	0.8902	103	640: 100% 32/32 2.6it/s 12.2s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 2/2 3.2it/s 0.6s
	all	32	483	0.978	0.902	0.949 0.768
Epoch	GPU mem	box loss	cls loss	dfl loss	Instances	Size

